

Task 11: Principal Component Analysis (PCA) – Wine Dataset

Objective

Apply PCA to reduce dimensionality of the Wine dataset and visualize in 2D.

Process Summary

1. Dataset loaded and features separated.
2. Data standardized using StandardScaler.
3. PCA applied to transform high-dimensional data.
4. Explained variance plotted (Scree Plot).
5. Reduced to 2 Principal Components and visualized.

Key Results

- PCA reduces dimensionality while keeping maximum variance.
- First two components capture most of the information.
- Makes visualization and ML training easier.

Interview Questions & Detailed Answers

1) What is Dimensionality Reduction?

Dimensionality reduction is the process of reducing the number of input variables in a dataset while keeping the most important information. High-dimensional datasets are complex and slow to process. Dimensionality reduction simplifies the data, removes noise and redundancy, and improves model performance.

2) Why do we use PCA?

PCA is used because:

- It reduces dataset size while keeping maximum information.
- It removes multicollinearity (correlated features).
- It speeds up machine learning training.
- It helps visualize high-dimensional data in 2D or 3D.
- It improves model generalization by reducing noise.

3) What is Explained Variance?

Explained variance tells us how much information (variance) each principal component captures from the original dataset. Higher explained variance means the component is more important.

4) What are Principal Components?

Principal components are new features created from linear combinations of original features. They are:

- Uncorrelated with each other
- Ordered by importance
- Designed to capture maximum variance.

5) Why is Standardization required before PCA?

PCA is based on variance. If features have different scales (like age vs salary), large-scale features dominate the PCA results. Standardization ensures all features contribute equally.

6) What is a Scree Plot?

A Scree Plot shows cumulative explained variance vs number of components. It helps decide how many components to keep.

7) Advantages of PCA

- Reduces overfitting
- Improves speed and efficiency
- Removes correlated features
- Helps visualization

Conclusion

PCA successfully reduced the Wine dataset into 2 dimensions while preserving most variance, making the data easier to visualize and analyze.